전산화학 (2012년 1학기) **1차 시험 해답**

(총 6 문제, 60 점)

1. **(총 15점)**

(a) **(8점)** $1+\sqrt{3}i = 2\left(\frac{1}{2}+\frac{\sqrt{3}}{2}i\right) = 2\big(\cos(\pi/3)+i\sin(\pi/3)\big) = 2e^{i\pi/3}$ 이므로

$$\begin{aligned}\sqrt{1+\sqrt{3}i}+(1+2i)^2 = &= \left(2e^{i\pi/3}\right)^{1/2}+(1+2i)^2\\ &= \sqrt{2}e^{i\pi/6}+(1+2i)^2\\ &= \sqrt{2}\left(\cos\left(\frac{\pi}{6}\right)+i\sin\left(\frac{\pi}{6}\right)\right)+1+4i-4\\ &= \sqrt{2}\left(\frac{\sqrt{3}}{2}+\frac{1}{2}i\right)-3+4i\\ &= \left(\frac{\sqrt{6}}{2}-3\right)+\left(\frac{\sqrt{2}}{2}+4\right)i.\end{aligned}$$

$\therefore \mathrm{Re}(z) = \frac{\sqrt{6}}{2}-3, \qquad \mathrm{Im}(z) = \frac{\sqrt{2}}{2}+4$ 이다.

(b) **(7점)**

$$z = 2^i = e^{i\ln 2} = \cos(\ln 2)+i\sin(\ln 2)$$

$\therefore \mathrm{Re}(z) = \cos(\ln 2), \qquad \mathrm{Im}(z) = \sin(\ln 2).$

2. **(5점)**

$$\begin{aligned}x &= r\sin\theta\cos\phi = 10\times\sin\left(\frac{\pi}{4}\right)\times\cos\left(\frac{3\pi}{4}\right) = 10\times\frac{\sqrt{2}}{2}\times-\frac{\sqrt{2}}{2} = -5\\ y &= r\sin\theta\sin\phi = 10\times\sin\left(\frac{\pi}{4}\right)\times\sin\left(\frac{3\pi}{4}\right) = 10\times\frac{\sqrt{2}}{2}\times\frac{\sqrt{2}}{2} = 5\\ z &= r\cos\theta = 10\times\cos\left(\frac{\pi}{4}\right) = 10\times\frac{\sqrt{2}}{2} = 5\sqrt{2}\end{aligned}$$

3. **(5점)**

$$\begin{aligned}r &= \sqrt{x^2+y^2+z^2} = \sqrt{24}\\ \theta &= \arccos\left(\frac{z}{r}\right) = \arccos\left(\frac{4}{\sqrt{24}}\right) = \arccos\left(\frac{2}{\sqrt{6}}\right)\\ \phi &= \arctan\left(\frac{y}{x}\right) = \arctan(1) = \frac{\pi}{4}\end{aligned}$$

4. **(10점)** Maclaurin 급수는 다음과 같이 주어진다.

$$f(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n = f(0)+f^{(1)}(0)x+\frac{1}{2}f^{(2)}(0)x^2+\cdots.$$

$$\begin{aligned}f'(x) &= (e^{-2x^2})' = -4xe^{-2x^2} \Rightarrow f'(0) = 0\\ f''(x) &= (e^{-2x^2})'' = -4e^{-2x^2}+16x^2e^{-2x^2} \Rightarrow f'(0) = -4\\ f'''(x) &= (e^{-2x^2})''' = 48xe^{-2x^2}-64x^3e^{-2x^2} \Rightarrow f'(0) = 0\\ f^{(4)}(x) &= \frac{d^4}{dx^4}(e^{-2x^2}) = 48e^{-2x^2}-384x^2e^{-2x^2}+256x^4e^{-2x^2} \Rightarrow f'(0) = 48\end{aligned}$$

그러므로

$$f(x) = e^{-2x^2} = 1 - 2x^2 + 2x^4 + \cdots \tag{1}$$

$x = 0.1$ 을 (1) 식에 대입하면

$$f(0.1) = 1 - 2(0.1)^2 + 2(0.1)^4 + \cdots \approx 0.980 \tag{2}$$

5. **(10 점)** 직교 좌표계에서의 한 점 (x, y) 는 극 좌표계에서 다음과 같이 변형된다.

$$x = r\cos\theta, \tag{3}$$
$$y = r\sin\theta.. \tag{4}$$

그러므로,

$$r^2 = x^2 + y^2, \tag{5}$$
$$\tan\theta = \frac{x}{y}. \tag{6}$$

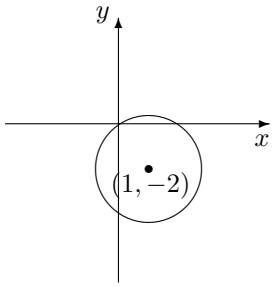
$r = 2\cos\theta - 4\sin\theta$ 를 다시 쓰면

$$r = 2\cos\theta - 4\sin\theta \Rightarrow r^2 = 2r\cos\theta - 4r\sin\theta.$$

(3), (4), (5) 식을 이용하면, 다음과 같은 식을 얻을 수 있다.

$$\begin{aligned} x^2 + y^2 = 2x - 4y &\Rightarrow x^2 - 2x + y^2 + 4y = 0, \\ &\Rightarrow x^2 - 2x + 1 - 1 + y^2 + 4y + 4 - 4 = 0, \\ &\Rightarrow (x-1)^2 + (y+2)^2 = (\sqrt{5})^2. \end{aligned} \tag{7}$$

(7) 식은 중심이 $(1, -2)$ 이고, 반지름 $r = \sqrt{5}$ 인 원을 나타낸다.



6. **(총 15 점)**

(a) **(총 5 점)** Euler 공식으로부터

$$e^{ib\theta} = \cos b\theta + i\sin b\theta, \tag{8a}$$
$$e^{-ib\theta} = \cos b\theta - i\sin b\theta, \tag{8b}$$

이므로, (8a) + (8b) $\Rightarrow 2\cos b\theta = e^{ib\theta} + e^{-ib\theta} \Rightarrow \cos b\theta = \frac{1}{2}(e^{ib\theta} + e^{-ib\theta})$.

(b) **(총 10 점)**

$$I = \int_0^\infty e^{-ax}\cos b dx \qquad (a > 0)$$

(a) 로부터

$$
\begin{aligned}
I &= \int_0^\infty e^{-ax}\frac{1}{2}\big(e^{ibx}+e^{-ibx}\big)dx \\
&= \frac{1}{2}\int_0^\infty \big(e^{-(a-ib)x}+e^{-(a+ib)x}\big)dx \\
&= \frac{1}{2}\left\{\Big[-\frac{1}{a-bi}e^{-(a-bi)x}\Big]_0^\infty + \Big[-\frac{1}{a+bi}e^{-(a+bi)x}\Big]_0^\infty\right\} \\
&= \frac{1}{2}\left(\frac{1}{a-bi}+\frac{1}{a+bi}\right) \\
&= \frac{1}{2}\frac{2a}{a^2+b^2} \\
&= \frac{a}{a^2+b^2}.
\end{aligned}
$$